

Technical Document #41: Blockchain - Comprehensive Overview

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Proof of Stake (PoS) selects validators based on their stake in the network. It is more energy-efficient than Proof of Work.

Proof of Work (PoW) is a consensus mechanism where miners solve complex puzzles to validate transactions. It is energy-intensive but secure.

Tokenomics studies the economic systems of cryptocurrency tokens. It covers supply mechanisms, incentive structures, and value capture.

Non-fungible tokens (NFTs) represent unique digital assets on the blockchain. They have transformed digital ownership and art markets.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Key Takeaways:

- Zero-knowledge proofs allow one party to prove knowledge of information without revealing the information itself.
- Smart contracts are self-executing contracts with terms directly written into code.
- Non-fungible tokens (NFTs) represent unique digital assets on the blockchain.